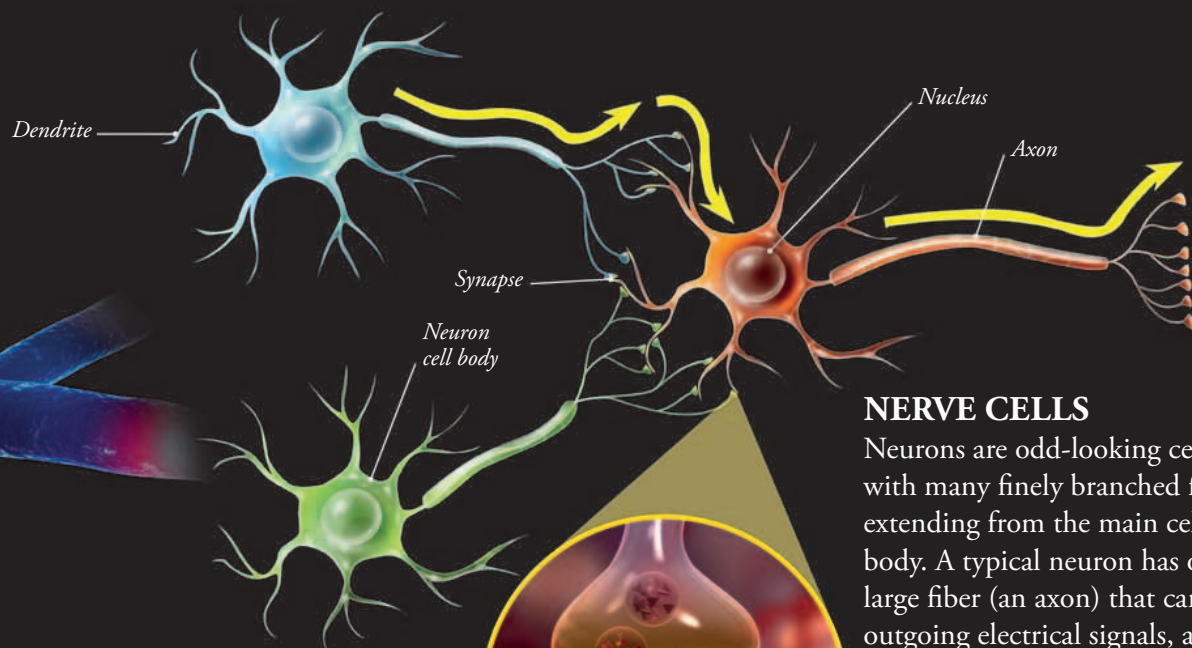
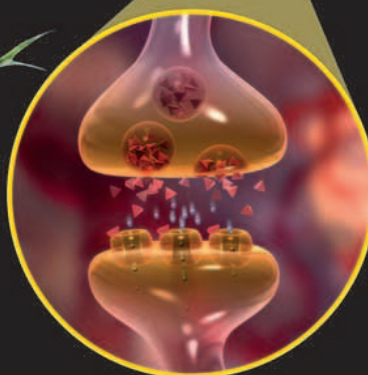




NERVE CELLS

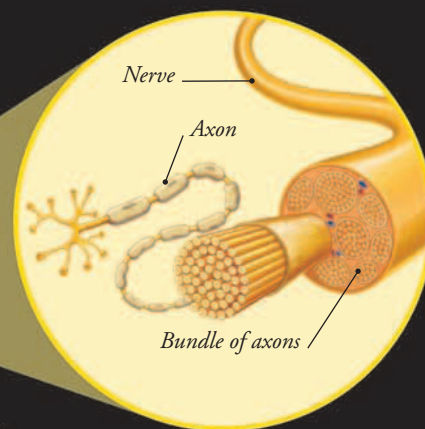
Neurons are odd-looking cells, with many finely branched fibers extending from the main cell body. A typical neuron has one large fiber (an axon) that carries outgoing electrical signals, and a large number of smaller fibers (dendrites) that carry incoming signals. Neurons connect with each other at junctions called synapses, but there's a tiny gap in the synapse that signals cannot jump across. Instead, chemicals called neurotransmitters flood into the gap and trigger a new signal in the next neuron.

▲ **CHEMICAL TRIGGER** *When the electrical signal reaches the end of a neuron, the cell releases chemicals that cross the gap (synapse) and switch on the next cell.*



▼ NERVES

Nerves are the body's main electric cables. A nerve is made up of bundles of long axons, each of which carries a separate electric signal. Tough casings protect the axons from damage.



BRANCHING OUT

Your body is riddled with nerves. Most nerves branch out from your spinal cord, which runs down the spine from the base of your brain. The nerves divide into finer and finer branches as they reach out into the body, eventually connecting with every single muscle cell and sensory cell.

